

[GETNET]ROTEIRO DE DRP PNR PRD: STANDBY READ-WRITE COM FLASHBACK

1) [STANDBY] Habilitar FRA

```
./db_profile
sqlplus /nolog
connect /as sysdba
```

```
$ df -g
```

Filesystem	GB blocks	Free	%Used	lused	%lused	Mounted on
/dev/lv_u02	2045.00	1472.12	29%	23117	1%	/u02

GROUP_NUMBER	NAME	SECTOR_SIZE	LOGICAL_SECTOR_SIZE	BLOCK_SIZE	ALLOCATION_UNIT_SIZE
1	ARCH_PARPRD	512	512	4096	4194304
2	DATA_AUX_PARPRD	512	512	4096	4194304
3	DATA_PARPRD	512	512	4096	4194304
4	DATA_REP_PARPRD	512	512	4096	4194304

ALLOCATION_UNIT_SIZE	STATE	TYPE	TOTAL_MB	FREE_MB	HOT_USED_MB	COLD_USED_MB	REQUIRE
4194304	CONNECTED	EXTERN	442368	435032	0	7336	
4194304	CONNECTED	EXTERN	147456	115584	0	31872	
4194304	CONNECTED	EXTERN	7667712	2995216	0	4672496	
4194304	CONNECTED	EXTERN	11010048	2628660	0	8381388	

B	REQUIRED_MIRROR_FREE_MB	USABLE_FILE_MB	OFFLINE_DISKS	COMPATIBILITY	DATABASE_COMPATIBILITY
	0	435032	0	19.0.0.0.0	10.1.0.0.0
	0	115584	0	19.0.0.0.0	10.1.0.0.0
	0	2995216	0	19.0.0.0.0	10.1.0.0.0
	0	2628660	0	19.0.0.0.0	10.1.0.0.0

V	CON_ID
N	0
N	0
N	0
N	0

```
log_archive_dest_1 string LOCATION=+ARCH_PARPRD
alter system set db_recovery_file_dest_size=1200G scope=both sid='*';
alter system set db_recovery_file_dest='/u02/oracle/backup/flashback' scope=both sid='*';
SQL> sho parameter recover
```

2) [PRIMARY|STANDBY] Desabilitar scripts de backup/purge de archivelogs

[PRIMARY]

Comentar linha de chamada de script de backup de archivelogs em PARPRD_GNCASNPA04698_ARCH_DCS_PROD.rman:

```
$ cd /u02/scripts/backup/rman
$ cd /u02/scripts/backup/rman
connect target /
connect catalog rmancat/rmancat@RCAT
#RUN { EXECUTE GLOBAL SCRIPT ARCH_DCS_PROD ; }
#RUN { EXECUTE GLOBAL SCRIPT MAINT_DCS_PROD ; }
```

[STANDBY] Desabilitar a limpeza dos archives

Comentar script de limpeza na crontab:

Clear Archive Logs DB PRIMARY

```
#57,27 * * * * /u02/scripts/clear_archives_gncasspa04699.sh
```

3) [PRIMARY] Parar OGG processes (incluindo MGR)

```
GGSCI> stop ER *
GGSCI> stop MGR
GGSC> info *
```

— Mapeamento processos OGG envolvidos: (avisar ao NOC sobre eventuais alertas de pumps que enviam dados para a [PRIMARY]):

```
[gncasnpa04698] EXT1PAR      schema: SVC_PNR > [gncasnpa04698] N/A          > [N/A] N/A schema: N/A
[gncasnpa04698] /ggs/dirdat/cc          > [gncasnpa04698] RMTTRAIL N/A          > [N/A]

[gncasnpa04698] EXT1PRN      schema: SVC_PNR > [gncasnpa04698] N/A          > [N/A] N/A schema: N/A
[gncasnpa04698] /ggs/dirdat/aa          > [gncasnpa04698] RMTTRAIL N/A          > [N/A]

[gncasnpa04698] EXT3PRN      schema: SVC_PNR > [gncasnpa04698] N/A          > [N/A] N/A schema: N/A
[gncasnpa04698] /ggs/dirdat/ea          > [gncasnpa04698] RMTTRAIL N/A          > [N/A]

[gncasnpa04698] EXT4PRN      schema: SVC_PNR > [gncasnpa04698] N/A          > [N/A] N/A schema: N/A
[gncasnpa04698] /ggs/dirdat/eb          > [gncasnpa04698] RMTTRAIL N/A          > [N/A]

[gncasnpa04698] EXT5PNR      schema: SVC_PNR > [gncasnpa04698] N/A          > [N/A] N/A schema: N/A
[gncasnpa04698] /ggs/dirdat/ca          > [gncasnpa04698] RMTTRAIL N/A          > [N/A]

--

[gncasnpa04698] EXTMGM1      schema: SVC_PNR > [gncasnpa04698] PXTMGM1          > [GNCASHPW03061] REPPNR1|REPPNR2|R
EPPNR3|REPPNR4 schema: dbo
[gncasnpa04698] /ggs/dirdat/ki          > [gncasnpa04698] RMTTRAIL F:\ggs\dirdat\kj > [GNCASHPW03061]

--

[gncasnpa04698] EXTPNR2      schema: SVC_PNR > [gncasnpa04698] PXTPNR2          > [GNCASSPA01354] REPPNR2 schema: PRD_
RTG_EXTR_ELETRONICO
[gncasnpa04698] /ggs/dirdat/ka          > [gncasnpa04698] RMTTRAIL /ggs/dirdat/kb > [GNCASSPA01354]
```

[gncasnpa04698] EXTPNR3 schema: SVC_PNR > [gncasnpa04698] PXTPNR3 > [gncasspl01262] REPPNR3 schema: SVC_PNR

[gncasnpa04698] /ggs/dirdat/kc > [gncasnpa04698] RMTTRAIL /ggs/dirdat/kd > [gncasspl01262]

--

[gncasnpa04698] N/A schema: SVC_PNR > [gncasnpa04698] PXT1PAR > [gncasspl01262] REPPAR1 schema: SVC_PAR

[gncasnpa04698] Redo Logs > [gncasnpa04698] RMTTRAIL /ggs/dirdat/cd > [gncasspl01262]

[gncasnpa04698] N/A schema: SVC_PNR > [gncasnpa04698] PXT3PRN > [GNCASSPA01354] REP1PNR schema: PRD_RT
G_EXTR_ELETRONICO

[gncasnpa04698] Redo Logs > [gncasnpa04698] RMTTRAIL /ggs/dirdat/ee > [GNCASSPA01354]

[gncasnpa04698] N/A schema: SVC_PNR > [gncasnpa04698] PXT4PRN > [GNCASNPA01040] REP1PNR schema: BHIST_P
NR

[gncasnpa04698] Redo Logs > [gncasnpa04698] RMTTRAIL /ggs/dirdat/ep > [GNCASNPA01040]

[gncasnpa04698] N/A schema: SVC_PNR > [gncasnpa04698] PXT5PNR > [gncasspl01262] REPPNR5 schema: SVC_PNR

[gncasnpa04698] Redo Logs > [gncasnpa04698] RMTTRAIL /ggs/dirdat/cb > [gncasspl01262]

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[gncasspa00548] N/A schema: N/A > [gncasspa00548] PXT2PRN > [gncasnpa04698] REP1PRN schema: SVC_PNR

[gncasspa00548] Redo Logs > [gncasspa00548] RMTTRAIL /ggs/dirdat/mc > [gncasnpa04698]

[gncasspa00548] N/A schema: N/A > [gncasspa00548] PXT1PRN > [gncasnpa04698] REP2PRN schema: SVC_PNR

[gncasspa00548] Redo Logs > [gncasspa00548] RMTTRAIL /ggs/dirdat/pr > [gncasnpa04698]

[gncasspa00548] N/A schema: N/A > [gncasspa00548] PXT3PRN > [gncasnpa04698] REP3PRN schema: SVC_PNR

[gncasspa00548] Redo Logs > [gncasspa00548] RMTTRAIL /ggs/dirdat/sl > [gncasnpa04698]

[gncasspa00548] N/A schema: N/A > [gncasspa00548] PXT1PRN > [gncasnpa04698] REP4PRN schema: SVC_PNR

[gncasspa00548] Redo Logs > [gncasspa00548] RMTTRAIL /ggs/dirdat/pr > [gncasnpa04698]

[GNCASHPW03061] EXTSQL01 schema: dbo > [GNCASHPW03061] PUMSQL01 > [gncasnpa04698] REPSQL01 schema: SV
C_PNR

```
[GNCASHPW03061] Redo Logs > [GNCASHPW03061] RMTTRAIL /qgs/dirdat/kg > [gncasnpa04698]
```

[STANDBY]

[PRIMARY]

[STANDBY]

5) [OPCIONAL][PRIMARY] Desabilitar redo log shipping do PRIMARY

IK: N/A (opcional) porque o DB [PRIMARY] será parado para esta simulação de DRP.

6) [OPCIONAL] [PRIMARY] Desabilitar database STANDBY

IK: (opcional). SE entretanto NÃO se optar pela desabilitação, o MRP terá que ser desativado manualmente em algumas etapas posteriores deste plano.

7) [PRIMARY] Parar DB

8) [STANDBY] Ativar flashback db

IK: ver observação item 6 deste roteiro: (em 15/07/2023, STANDBY NÃO foi desabilitado no DG BROKER):

```
alter database recover managed standby database cancel;
alter database flashback on;
create restore point RP_PARPRDDCS_DRP_20230715 guarantee flashback database;
```

9) [STANDBY] Procedimento para abrir em R/W o então STANDBY

```
SQL> alter database activate standby database;
--startup mount force; --<-- verificar se pode ser feito pelo srvctl (para o caso de RAC Databases)
$ srvctl stop database -d PARPRDDCS
$ srvctl start database -d PARPRDDCS -o mount --<-- ATENÇÃO que com este comando, associado ao DG BROKER ativo, inicia automaticamen
te o MRP!
-- N/A alter database set standby database to maximize performance; --<== N/A: O db já está em MAXIMUM PERFORMANCE.
```

[PRIMARY | STANDBY] Verificar database role, open_mode e status das db instances:

```
set lines 2000
set trimspool on
set pagesize 1000
alter session set nls_date_format='dd/mm/yyyy hh24:mi';
select dbid,name,db_unique_name,primary_db_unique_name,
       created,resetlogs_time,log_mode,
       open_resetlogs,open_mode,database_role,dataguard_broker,
       force_logging,platform_name,switchover_status,
       protection_mode, protection_level
from v$database;
select * from gv$instance;
```

On EACH node of RAC cluster:

```
SQL> alter database open;
```

10) [STANDBY] Subir db service PARPRDSRV

```
$ srvctl status service -d PARPRDDCS -s PARPRDSRV

Service PARPRDSRV is not running.

$ srvctl start service -d PARPRDDCS -s PARPRDSRV

$ srvctl status service -d PARPRDDCS -s PARPRDSRV

Service PARPRDSRV is running
```

as gi o.s. user (grid):

verificar se listener está com o db service PARPRDSRV

```
$ lsnrctl status LISTENER
```

11) [STANDBY] Testes aplicativos, criação de datafiles, etc.

Resp. equipe APP.

12) [STANDBY] Baixar db service PARPRDSRV

as rdbms/db o.s. user (oracle):

```
$ srvctl status service -d PARPRDDCS -s PARPRDSRV

Service PARPRDSRV is running.

$ srvctl stop service -d PARPRDDCS -s PARPRDSRV

$ srvctl status service -d PARPRDDCS -s PARPRDSRV

Service PARPRDSRV is not running
```

as gi o.s. user (grid):

verificar se listener está com o db service PARPRDSRV

```
$ lsnrctl status LISTENER
```

13) [STANDBY] Retorno do db R/W para a condição STANDBY ("ADG")

[STANDBY]

```
./db_profile
sqlplus /nolog
connect /as sysdba
-- startup mount force;
$ srvctl stop database -d PARPRDDCS
$ srvctl start database -d PARPRDDCS -o mount --<-- ATENÇÃO que com este comando, associado ao DG BROKER ativo, inicia automaticament
e o MRP!
SQL> flashback database to restore point RP_PARPRDDCS_DRP_20230715;
SQL> alter database convert to physical standby
SQL> flashback database to restore point RP_PARPRDDCS_DRP_20230715;
SQL> alter database convert to physical standby;
```

— Restart MRP:

```
alter database recover managed standby database disconnect;
```

—Cleanup:

—Depois de [STANDBY] resincronizado:

— Drop restore point:

```
SQL> drop restore point RP_PARPRDDCS_DRP_20230715;
```

— Desligar flashback db: (este cmd apaga os flashback logs correlatos)

```
alter database flashback off;
```

— Desabilitar FRA:

```
alter system set db_recovery_file_dest_size = '1' scope=both sid="";
alter system set db_recovery_file_dest = " scope=both sid="";
SQL> sho parameter recover
```

14) [PRIMARY] Reiniciar DB

```
$ srvctl start database -d PARPRDDCN
```

15) [PRIMARY] Reiniciar OGG processes (incluindo MGR)

```
GGSCI> start MGR
GGSCI>start ER *
```

16) [PRIMARY] Acompanhar, verificar e validar fluxos OGG

IK: Se basear no mapeamento OGG constante no item 3 deste roteiro.

17) [OPCIONAL] [PRIMARY] Reabilitar redo log shipping do PRIMARY

IK: (N/A) SE redo log shipping tiver sido desabilitado anteriormente (ref. item 5 deste roteiro).

```
DGMGRL> show configuration verbose
Configuration - DR_PAR
Protection Mode: MaxPerformance
Members:
PARPRDDCN - Primary database
PARPRDDCS - Physical standby database
```

18) [OPCIONAL] [PRIMARY] Reabilitar configuração do DB STANDBY no DG BROKER:

IK: (N/A) SE configuração do STANDBY tiver sido desabilitada anteriormente (ref. item 6 deste roteiro).

```
DGMGRL> connect /
Connected as SYSDBG.
DGMGRL> show configuration
DGMGRL> ENABLE DATABASE 'PARPRDDCS';
DGMGRL> show configuration
DGMGRL> show database verbose 'PARPRDDCS';
```

19) [STANDBY] Open STANDBY em READ ONLY RTA

```

./db_profile
sqlplus /nolog
connect /as sysdba
--startup nomount;
--alter database mount standby database;
--alter database open read only;
$ srvctl stop database -d PARPRDDCS
$ srvctl start database -d PARPRDDCS --<-- ATENÇÃO que com este comando, associado ao DG BROKER ativo, inicia automaticamente o MRP!
-- alter database recover managed standby database using current logfile disconnect noparallel;
-- alter database recover managed standby database using current logfile disconnect from session;

```

20) [STANDBY] Verificações

— [PRIMARY | STANDBY] Verificar database role, open_mode e status das db instances:

```

set lines 2000
set trimspool on
set pagesize 1000
alter session set nls_date_format='dd/mm/yyyy hh24:mi';
select dbid,name,db_unique_name,primary_db_unique_name,
       created,resetlogs_time,log_mode,
       open_resetlogs,open_mode,database_role,dataguard_broker,
       force_logging,platform_name,switchover_status,
       protection_mode, protection_level
from   v$database;
select * from gv$instance;

```

— [PRIMARY | STANDBY] Parâmetros spfile ADG e de listener:

```

set lines 2000
set trimspool on
set pagesize 1000
col name          format a30
col value         format a60
col display_value format a30
col description   format a50
col update_comment format a30
alter session set nls_date_format='dd/mm/yyyy hh24:mi';

```

— Parâmetros ADG:

```

select sysdate refdate,d.name db_name,
       p1.name,p1.inst_id,p1.value,p1.isdefault,p1.issys_modifiable,p1.isinstance_modifiable
from   gv$parameter p1,v$database d
where  p1.name in ('db_create_file_dest',
                  'db_file_name_convert',
                  'db_recovery_file_dest',
                  'db_recovery_file_dest_size',
                  'db_unique_name',
                  'dg_broker_start',
                  'dg_broker_config_file1',
                  'dg_broker_config_file2',
                  'fal_client',
                  'fal_server',
                  'local_listener',
                  'log_archive_config',
                  'log_archive_dest_1',
                  'log_archive_dest_2',
                  'log_archive_dest_3',
                  'log_archive_dest_state_1',
                  'log_archive_dest_state_2',
                  'log_archive_dest_state_3',
                  'log_archive_format',
                  'log_archive_max_processes',
                  'log_file_name_convert',
                  'remote_login_passwordfile',
                  'standby_file_management')
order by p1.name,p1.inst_id;

```

— Parâmetros listener:

— [PRIMARY] Ver última seq gerada:

— [STANDBY]

— Query IK/PS de 02/10/2017:

—XXX—

— [STANDBY] Verificar se MRP está em execução:

— [PRIMARY|STANDBY]


```
col dest_name format a50
select  ad.inst_id
,      ds.dest_id id
,      ad.dest_name
,      ad.destination
,      ad.error
,      ad.status
,      ds.database_mode db_mode
,      ad.archiver type
,      ds.recovery_mode
,      ds.protection_mode
,      ds.standby_logfile_count "SRLs"
,      ds.standby_logfile_active active
,      ds.archived_seq#
,      ad.max_failure
,      ad.db_unique_name
,      ad.log_sequence
from    gv$archive_dest_status ds
,      gv$archive_dest ad
where   ds.dest_id = ad.dest_id
and     ds.inst_id = ad.inst_id
and     ad.status != 'INACTIVE'
order by ds.dest_id;
```

—xxx—

— [PRIMARY|STANDBY]

```
alter session set nls_date_format='dd/mm/yyyy hh24:mi:ss';
set lines 2000
set trimspool on
col dest_id format 99999
select * from gv$dataguard_status order by timestamp,inst_id;
```

— MONITORAMENTO FRA (flash recovery area):

```
archive log list
sho parameter recover
select  frau.file_type                                as type,
        frau.percent_space_used/100 * rfd.space_limit /1024/1024/1024    "USED_GB",
        frau.percent_space_used                                           "USED%",
        frau.percent_space_reclaimable                                    "RECLAIMABLE%",
        frau.percent_space_reclaimable/100 * rfd.space_limit /1024/1024/1024 "RECLAIM_GB",
        frau.number_of_files                                              "FILES#"
from     v$flash_recovery_area_usage frau,
        v$recovery_file_dest rfd
order by file_type;                                "FILES#"
```

— DG BROKER:

— [PRIMARY]

```
$ dgmgrl
DGMGRL> connect /
DGMGRL> show configuration
DGMGRL> show database verbose 'PARPRDDCS';
DGMGRL>
```

21) Reabilitar backup de archivelogs

— [PRIMARY] Reabilitar script de backup/purge de archives

— Descomentar linha de chamada de script de backup de archivelogs em PARPRD_GNCASNPA04698_ARCH_DCS_PROD.rman:

```
cd /u02/scripts/backup/rman
connect target /
connect catalog rmancat/rmancat@RCAT
#RUN { EXECUTE GLOBAL SCRIPT ARCH_DCS_PROD ; }
#RUN { EXECUTE GLOBAL SCRIPT MAINT_DCS_PROD ; }
```

— [STANDBY] Reabilitar a limpeza dos archives

— Descomentar script de limpeza na crontab:

Clear Archive Logs DB PRIMARY

#57,27 * * * * /u02/scripts/clear_archives_gncasspa04699.sh